Dominik Hüppin

MSc ETH in Physics

Website · LinkedIn · Zürich, Switzerland

Physicist with experience in numerical modelling, electromagnetic simulation, data analysis, Python-based automation, and experimental research. International research experience at ETH Zürich, Kyoto University, and ISAS/JAXA.

Education

07/2025–09/2026 **ETH Zürich** · MSc Physics GPA 5.44/6.0 — Zürich, Switzerland

Focus: Astrophysics and Quantum Field Theory

09/2020–07/2025 **ETH Zürich** · BSc Physics GPA 5.0/6.0 — Zürich, Switzerland

Focus: Quantum Mechanics and Astrophysics

Research & Experimental Experience

04/2026–09/2026 **Sirin Orbital Systems AG / ETH Zürich / Kyoto University** · Master's Thesis (Grade: 6.0/6.0, nominated for the ETH Medal) — Zürich, Switzerland & Kyoto, Japan

- Designed, simulated, fabricated, and experimentally characterized a 61 GHz mechanically beam-steerable travelling-wave slot-array antenna for wireless power transfer.
- Combined antenna design and electromagnetic simulation with prototype fabrication and experimental validation.

10/2025 **Institute of Space and Astronautical Science (ISAS/JAXA)** · Skills Acquisition Programme — Sagamihara, Japan

- Worked in Prof. Koji Tanaka's laboratory on microwave wireless power transfer for space applications.
- Designed and simulated a 5.8 GHz antenna array for space applications.

09/2025 **Kyoto University, Research Institute for Sustainable Humanosphere (RISH)** · Semester Project — Kyoto, Japan

- Designed, simulated, fabricated, and experimentally characterized a microwave RF-to-DC rectifier for wireless power transfer in Prof. Naoki Shinohara's laboratory.
- Optimized RF-to-DC conversion efficiency through iterative simulation, fabrication, and measurement.

02/2024–06/2024 **ETH Zürich, Hybrid Quantum Systems Group** · Research Project — Zürich, Switzerland

- Developed Python-based automation for data acquisition in single-photon transduction experiments.
- Implemented automated measurements and histogram generation, improving measurement efficiency and reproducibility.

Skills & Languages

Programming: Python, C++, MATLAB, LaTeX

Simulation: CST Studio Suite, Advanced Design System (ADS)

Software: Microsoft Office

Languages: German (native), English (C1), Spanish (B2), French (basic)

Personal Interests

Photography, Football, Endurance Sports, Surfing